

strip has dimension exactly D for every choice of nonzero weights (Theorem 94); the two-vertex, path and cycle reductions are special cases. Every claim below is labelled by the standard it meets. Four standards appear, in decreasing strength: *proved*, meaning a proof in the text; *proved by an exact certificate*, meaning an explicit matrix whose nullity is computed in $\mathbb{Z}$ or $\mathbb{Q}(\zeta_n)$; *proved by interval certification*, meaning a Krawczyk contraction bound establishing that a true real solution exists in an explicit box; and *numerical*, meaning a floating-point computation only. We also list what we have refuted, including our own earlier claims.

- **Proved.** Theorems 36, 40 and Theorem 44 with Corollary 45; Propositions 46 and 47; Theorem 63; Theorem 64 and Theorem 60, the nullity classifications of the adjacency matrix and of its signed variant; Theorem 49, that $c^2 = 0$ if and only if the symbol factors as $(s+a)(L_k+t)$, with Corollaries 50 and 51 and Proposition 52; Theorem 62, that no rational period-one symbol attains 14 at $k = 6$ in either matrix class; Theorem 72, that a period-one certificate covers exactly one divisibility class, which contains at most one prime, together with the identity $e_2 = -k$ for every $k \geq 3$; Theorem 65 — under the hypothesis $\det M(\zeta) \neq 0$, which an earlier version omitted — with Theorem 66 for cyclic covers; Theorems 80, 81, 82 and 83; the two refutations, Theorems 85 and 129; and Propositions 86 and 88.
- **Proved by an exact certificate.** $Z(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, and for every n divisible by 7 (Theorem 53); $Z(P(n, 3)) = 8$ for every n divisible by 10 (Theorem 58); $Z(P(n, 4)) = 10$ for every n divisible by 60, 70 or 90; $Z(P(n, 5)) = 12$ for every n divisible by 24; $Z(P(n, 7)) = 16$ for every n divisible by 120 (Theorem 61). The $k = 4, 5$ values were the first exact values for $k \geq 4$; the $k = 7$ value replaces a floating-point claim we previously made at a single n . In the $k = 2$ ($7 \mid n$), $k = 3$ and $k = 7$ cases the certificate is an integer matrix with entries in $\{0, \pm 1\}$.
- **Proved by exact certification.** $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$, and $Z(P(n, 4)) = M(P(n, 4)) = 10$ for every $n \geq 29$ (Theorem 109), and $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$ (Theorem 126), from Theorem 107 together with certified identity tiles of every length in $[21, 44)$ at $k = 3$ and $[29, 62)$ at $k = 4$, plus Theorem 100 for $17 \leq n \leq 20$; at $k = 5$ the certified lengths are $[54, 73) \cup \{74, 75\}$, which do not form an interval $[L, 2L)$ but generate every $n \geq 162$ as a sum. No congruence condition appears and the primes are included. All 77 tile certifications — 23 at $k = 3$, 33 at $k = 4$, 21 at $k = 5$ — and all 17 per- n certifications are Krawczyk contraction bounds with $\alpha \leq 1.0 \times 10^{-9}$, each evaluated in *exact rational arithmetic* (Section 20); no step rests on a floating-point nullity. Each rests on a Krawczyk contraction bound for the bilinear system $A(w)K(X) = 0$, with the square subsystem chosen structurally so that it is equivalent to $AK = 0$ rather than merely a maximal-rank selection of it. By Theorem 72 no period-one certificate can reach these n at all, so this is the only route to them that we know.
- **Proved (finite).** Every tabulated value of Z , by exhaustive search over vertex subsets. Combined with the above this gives Theorem 110: $Z(P(n, k))$ is determined for every admissible n when $k = 2$, $k = 3$ and $k = 4$, with no value left open in any of the three rows. In particular Theorem 111, $Z(P(n, 3)) = 8$ for every $n \geq 13$ with 13 optimal: the four values $n = 13, 14, 15, 16$ are exhaustive searches and the rest is Theorem 109, so the stabilization threshold at $k = 3$ — recorded as open in [7] — is exactly 13. This is the one result here whose statement needs no matrix theory to check.
- **Numerical, not exact.** Theorem 76: $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$ and $Z(P(96, 8)) = 18$. The certificates are period-three matrices whose weights come from a

Newton solve, verified on the full $2n \times 2n$ matrix with a wide spectral separation and every required entry a substantial fraction of the matrix scale, but not in exact arithmetic. We do not claim these at the standard of the items above. ($Z(P(96, 7)) = 16$ was in this category and is now exact.)

- **Observation supported by computation, not proved.** The rank law $\text{rank} = \min(2d + 1, k + 1)$, verified for $3 \leq k \leq 13$ and $1 \leq d \leq 7$. Also the saturation of the reachable set of monodromies, at dimension 31 for $k = 2$ and 51 for $k = 3$.
- **Conjectural.** Conjecture 41; and Conjecture 103 for $k \geq 6$. The leading-coefficient criterion, a conjecture in earlier versions, is now Theorem 91; what remains open there is the case of a top coefficient with two or more monomials, such as the tie $t_m = p + q$ of Theorem 98. For $k = 3$ and $k = 4$ the conjecture is a theorem (Theorem 109), and for $k = 5$ it is a theorem with a non-optimal threshold (Theorem 126, $N(5) \leq 162$); what remains open is whether $N(k)$ exists for every k , for which we prove only the necessary condition (4).
- **Refuted (ours).** The regular-base ceiling conjectured in Remark 89, and with it the conjecture of an earlier version that $M(B^n) = 6$ on the K_4 family of Section 21, which would have given $Z - M \rightarrow \infty$ on cubic graphs: Theorem 129 gives $M(B^n) \geq n$. The accompanying statement that the *equivariant* maximum nullity of that family is 6 was also false, because Theorem 65 needs $\det M(\zeta) \neq 0$ and a rank-one fibre makes it vanish identically. The conjecture had been written into the paper, the board and the interview script as the strongest result in the project.
- **Refuted (ours).** That the period-one construction “never exceeds 6 for every $k \geq 2$ ”. Three roots can be prescribed and no more, but the remaining $k - 2$ can land on the grid by Galois conjugacy, and do. The period-two construction we built to get past 6 was unnecessary and its values are superseded.
- **Refuted (ours).** That $k = 3$ and $k = 7$ admit no rational period-one symbol attaining $2k + 2$. That enumeration imposed $c^2 > 0$, the condition for a *symmetric* certificate, where $c^2 \neq 0$ suffices: c^2 is the product of the two spoke weights and only its vanishing is fatal. Six symbols at $k = 3$ and one at $k = 7$ survive, giving Theorems 58 and 61.
- **Refuted (ours).** The unconditional bound $Z(B^n) \geq D$ for cyclic covers, by $Z(P(8, 2)) = 5 < 6$. Only the conditional form $Z(B^n) \geq \nu$ for an attained ν survives.
- **Refuted (ours).** The single-threshold form $n_0(k) = 5k - 2$, by $Z(P(29, 6)) = 12$; and any bound of the form $Z \leq 2(\gcd(n, j) + \gcd(n, k))$, by $Z(P(11, 3)) = 7 > 4$.
- **Refuted (ours).** That the threshold for reaching the ceiling with a general matrix is $n \geq (2k + 2)^2/5$. Spending the gauge gives (4), $n \geq (k + 1)(2k + 3)/3$, which is larger; the earlier figure led us to test $k = 6$ at $n = 27, 29, 31$, all below the true threshold of 35.
- **Refuted (ours).** That $N(k)$ in Conjecture 103 is $O(k)$. The saturation data alone suggests $4k + 3$, but (4) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic.
- **Refined (ours).** We earlier recorded as refuted the claim that the monodromy is symplectic, on the grounds that the *standard* form is not preserved. That is true and was the wrong conclusion to draw. With the docking pattern fixed there is a unique invariant skew form B , and it is nondegenerate, so tile products are symplectic with respect to B (Observation 112).

The earlier statement — that the space of invariant skew forms for a single product has dimension $k + 1$, so eigenvalues come in reciprocal pairs — remains correct; it is the family, not the individual, that pins B down.

- **Observation, not proved.** Observations 112 and 113: the invariant form and the full-dimensionality of the reachable set, verified for $2 \leq k \leq 6$ but not proved. These are what reduce the existence of identity tiles for all k to a controllability question in $Sp(2k + 2, \mathbb{R})$, stated in Section 20. We do not claim that reduction settles the question: subsemigroups with nonempty interior in $Sp(2n, \mathbb{R})$ need not be the whole group.
- **Inconclusive, not negative (ours).** The per- n search of Section 20 at $k = 6$ returned nothing at $n = 41, 43, 45, 47, 49, 51$, and the tile search at $k = 5$ nothing at $41 \leq \ell \leq 45$ — the latter now explained: the first $k = 5$ tile is at $\ell = 54$, so that search was below the range where tiles exist. Both ran at a restart budget that we have since seen condemn a length that does solve: $\ell = 21$ at $k = 3$ stalled at 5.6×10^{-2} on four restarts and produced a tile at 1.4×10^{-14} on twenty-five. We therefore record these as searches we did not push far enough, not as evidence about $k = 5$ or $k = 6$. The one regularity we have found is that every length eventually solved first appeared at a residual of 10^{-2} or below, and the $k = 5$ and $k = 6$ attempts sit at 10^{-1} .
- **Withdrawn as unreliable (ours).** The eigenvalue-minimising search whose plateaus we previously reported as evidence that the ceiling is unreachable at various (n, k) . It gave maximum nullity 7 at $P(20, 3)$, where 8 is attained — by Theorem 58 and, in the symmetric class, by the method of Section 20. It collapses onto the degenerate strata (all spokes zero, or all edges zero). Its negative results should not be read as evidence, and the conclusion we drew from them, that exactness for $k \geq 6$ was unlikely, was not supported.
- **Open.** Conjecture 103 for $k \geq 6$, where we have no tiles at all and the count gives only $\ell \geq 54$. At $k = 5$ the conjecture now holds for $n \geq 162$ (Theorem 126) but not below it: we have 21 of the 54 lengths $[54, 108)$ would require, and $54 \leq n < 162$ is settled only at the 64 values that are sums of them. Also whether $M = Z$ in general, and the rank law. No (n, k) is proved to have $M(P(n, k)) < 2k + 2$, and the one case where we expected a strict gap, $(n, k) = (10, 2)$, turned out not to be one: $M(P(10, 2)) = Z(P(10, 2)) = 6$ by the integer certificate of Remark 55.
- **Superseded (ours).** The claim that the period- d method “cannot settle prime n at all” and that a different construction would be needed there. That is correct for the period- d method and is now a theorem (Theorem 72), but it is no longer a limitation of the paper: Section 20 settles prime n for $k = 3$.
- **Not determined.** Whether Theorem 40 is attained for $k \geq 3$; whether the degeneracy of Section 19 at $n = 48, 54$ is essential.

References

- [1] AIM Minimum Rank – Special Graphs Work Group, Zero forcing sets and the minimum rank of graphs, *Linear Algebra Appl.* **428** (2008) 1628–1648.
- [2] S. Fallat and L. Hogben, Variants on the minimum rank problem: a survey II, arXiv:1102.5142 (2011).

- [3] L. Duong, B. K. Kroschel, M. Riddell, K. N. Vander Meulen, A. Van Tuyl, Maximum nullity and zero forcing of circulant graphs, *Special Matrices* **8** (2020), no. 1, 221–234.
- [4] B. Brimkov, C. C. Fast, I. V. Hicks, Computational approaches for zero forcing and related problems, *European J. Oper. Res.* **273** (2019) 889–903; arXiv:1704.02065.
- [5] J. S. Fabila-Carrasco and F. Lledó, Covering graphs, magnetic spectral gaps and applications to polymers and nanoribbons, *Symmetry* **11** (2019) 1163.
- [6] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, Computing the zero forcing number for generalized Petersen graphs, *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020), 183–193.
- [7] A. Krishnan, A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$, arXiv:2607.19412 (2026).
- [8] A. Terras, *Zeta Functions and Chaos*, lecture notes, October 12, 2009; and *Zeta Functions of Graphs: A Stroll through the Garden*, Cambridge Studies in Advanced Mathematics 128, Cambridge University Press, 2010. The equivalence we use is stated there as Exercise 9: for a $(q + 1)$ -regular graph the Riemann Hypothesis holds if and only if every eigenvalue λ of the adjacency matrix with $|\lambda| \neq q + 1$ satisfies $|\lambda| \leq 2\sqrt{q}$.
- [9] S. Akbari, E. Vatandoost and Y. Golkhandy Pour, *Maximum nullity and zero forcing number on cubic graphs*, arXiv:1705.09773 (2017).
- [10] A. Lubotzky, R. Phillips and P. Sarnak, Ramanujan graphs, *Combinatorica* **8** (1988), 261–277.
- [11] R. Gera and P. Stănică, The spectrum of generalized Petersen graphs, *Australasian J. Combinatorics* **49** (2011), 39–45. Theorem 2.4 gives the quadratic $\delta^2 - (\alpha_j + \beta_j)\delta + \alpha_j\beta_j - 1 = 0$ with $\beta_j = 2T_k(\alpha_j/2)$, which is our Lemma 4.
- [12] T. A. Le and J. W. Sander, Integral circulant Ramanujan graphs of prime power order, *Electron. J. Combin.* **20** (2013), no. 3, #P35.
- [13] J. H. Conway and A. J. Jones, Trigonometric diophantine equations (On vanishing sums of roots of unity), *Acta Arith.* 30 (1976), 229–240.
- [14] I. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, Mathematical Association of America, 1956, Corollary 3.12.
- [15] A. Schrijver, *Combinatorial Optimization: Polyhedra and Efficiency*, Springer, Berlin, 2003 (Chapter 17, weighted bipartite matching and the assignment problem).

$$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2} |xy| + 3 \geq 0$$

one criterion, four fixed lenses, the same for every k

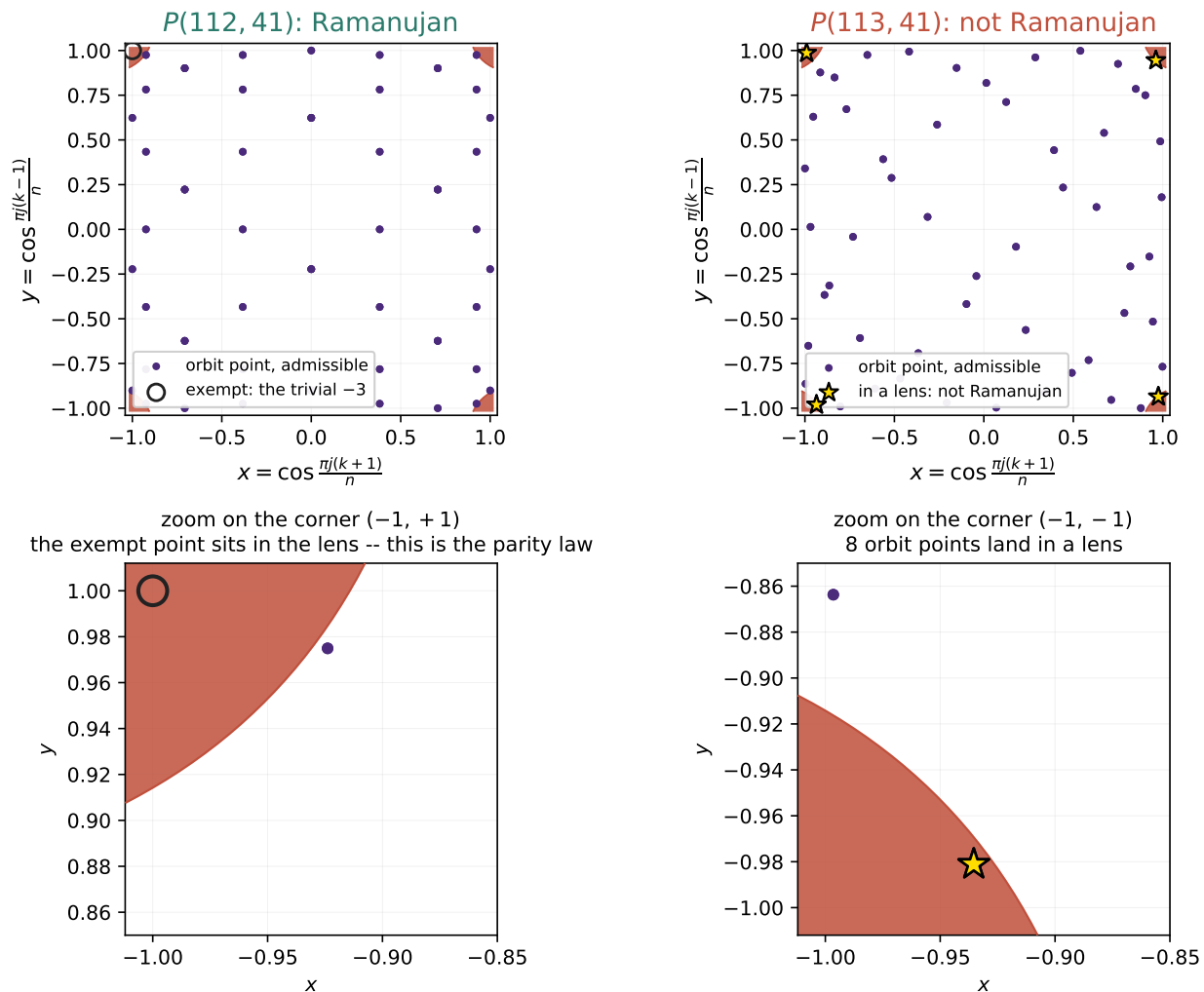


Figure 1: The corner criterion. Left, the largest Ramanujan member of the whole family; right, the cover one step larger, which fails. The four forbidden regions are the same in both pictures and for every k — only the map into the square changes. In the left-hand zoom the single point inside a region is the exempt trivial -3 , which is why bipartiteness helps.

k	B_k	count	the n for which $P(n, k)$ satisfies the RH
1	15	9	3–8, 10, 12, 14
2	23	19	5–23
3	32	18	7–16, 18, 20, 22, 24, 26, 28, 30, 32
4	42	34	9–42
5	51	28	11–26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50
6	61	20	13–16, 18, 20–23, 25, 27–28, 30, 32, 35, 37, 39, 42, 44, 49
7	71	39	15–36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70
8	81	16	17, 19–22, 24, 26, 28–29, 31, 33, 35, 38, 40, 42, 49
9	91	50	19–46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90
10	101	14	21, 23, 25–26, 28, 30, 32, 34, 37, 39, 41, 43, 50, 52
11	111	36	23–26, 28–32, 35–38, 40–42, 46–48, 50, 52–53, 58, 60, 62, 64, 70, 72, 74, 82, 84, 92, 94, 96, 104, 106
12	121	9	27, 29–30, 32, 34, 38, 40, 45, 51
13	131	23	28–30, 32, 34–37, 42–44, 46, 48–49, 56, 58, 60, 70, 72, 74, 84, 86, 98
14	141	7	31, 33, 35–36, 38, 46, 53
15	151	19	33–36, 38, 40–42, 49–50, 52, 54, 56, 66, 68, 70, 82, 84, 98
16	161	6	35, 39–40, 42, 44, 52
17	171	16	37–40, 42, 44, 46–47, 56, 58, 60, 62, 74, 76, 78, 94
18	181	3	45–46, 50
19	191	17	42–44, 46, 48, 50, 52–53, 64, 66, 68, 70, 84, 86, 88, 104, 106
20	201	3	45, 49–50
21	211	11	46, 48, 50, 54, 56, 58, 70, 72, 76, 92, 96
22	221	1	49
23	231	10	52–54, 56, 60, 62, 76, 78, 84, 106
24	241	1	53
25	251	8	56, 58, 60, 66, 68, 84, 90, 92
27	271	8	60, 62, 64, 70, 72, 74, 90, 98
29	291	9	64, 66, 68, 70, 76, 78, 96, 98, 106
31	311	4	70, 74, 82, 84
33	331	5	74, 76, 78, 86, 90
35	351	5	78, 80, 84, 92, 96
37	371	3	82, 84, 88
39	391	3	88, 102, 106
41	412	2	98, 112
43	432	2	96, 98
45	452	2	100, 108

No k outside this table admits any n : the list is finite in both variables.

Table 1: The complete classification, every n and every k . For each k , B_k is the per- k bound of Theorem 12; the rows stop at $k = 45$ because by Theorem 13 no larger k admits any n at all. Generated directly from the certification run.

k	bound $2k + 2$	least n attaining it	least n from which it holds always
2	6	7	9 (it fails at $n = 8$)
3	8	10	13 (it fails at $n = 11, 12$)
4	10	18	18 (Theorem 110)
5	12	23	not determined
6	14	28	not determined; $Z(P(29, 6)) = 12$

Table 2: The bound is attained well before it is attained *always*: at $k = 3$ it first holds at $n = 10$, fails again at $n = 11$ and 12 , and holds for every $n \geq 13$ (Theorem 111). The two columns are different quantities and conflating them is easy; at $k = 6$ the bound is attained at $n = 28$ and lost at $n = 29$, so no stabilization threshold is known there. At $k = 4$ the two columns happen to coincide at 18, which is not a pattern: they differ at $k = 2$ and $k = 3$. All values by exhaustive search.



Figure 2: The deficit $\delta^*(n)$, the best achievable nontrivial spectral radius at each size, measured from the Ramanujan threshold. Green points are sizes at which an optimal member exists; the last is $n = 112$. The curve rises to the asymptote $3 - 2\sqrt{2}$, which by Proposition 33 is the most that can ever be lost. The two branches are the parity law.

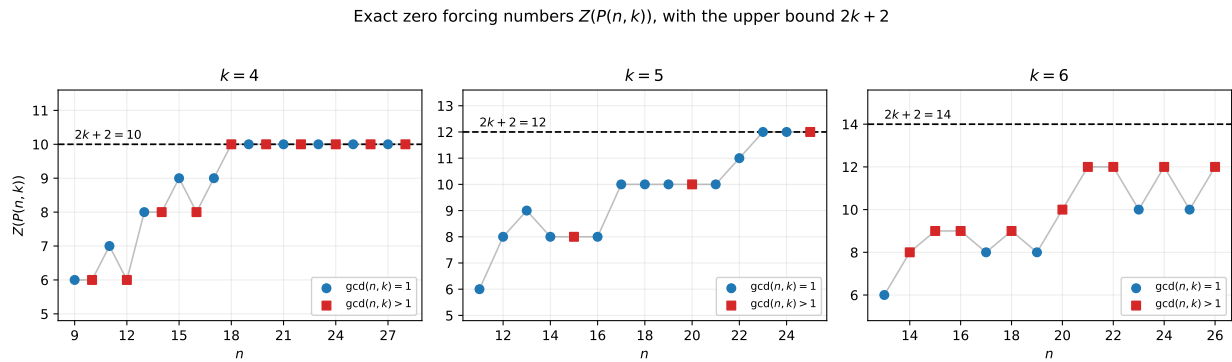


Figure 3: Exact $Z(P(n, k))$ for $k = 4, 5, 6$ against the bound $2k + 2$. Points are separated by whether $\gcd(n, k) = 1$. The sequences are non-monotone: $k = 4$ drops at $n = 12, 16$ and $k = 6$ at $n = 23, 25$.

k	$\mathcal{D}$	$\text{lcm}(\mathcal{D})$	F	a	d	c^2	nullity
2	$\{3, 10\}$	30	$sL_2 - 1$	0	0	1	$6 = 2k + 2$
2	$\{14\}$	14	Ψ_{14}	-1	0	1	$6 = 2k + 2$
3	$\{1, 4, 5\}$	20	$\Psi_1 \Psi_4 \Psi_5$	-1	-1	1	7
4	$\{4, 30\}$	60	$(s+1)L_4 - s - 2$	1	-1	1	$10 = 2k + 2$
4	$\{5, 14\}$	70	$\Psi_5 \Psi_{14}$	0	1	1	$10 = 2k + 2$
4	$\{5, 18\}$	90	$\Psi_5 \Psi_{18}$	1	0	1	$10 = 2k + 2$
5	$\{3, 6, 24\}$	24	$sL_5 - 1$	0	0	1	$12 = 2k + 2$

Table 3: Exact period-one certificates. The $k = 2$, $k = 4$ and $k = 5$ rows give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every n divisible by the stated modulus.

k	$2k + 2$	best period-one nullity	attained at
2	6	6	every n with $9 \leq n \leq 130$ except $n = 7, 8, 10$
3	8	8	$n = 40, 60, 80, 120$
4	10	10	$n = 60, 70, 90, 120$
5	12	12	$n = 24, 48, 72, 96, 120$
6	14	8	$n = 35, 37, 70, 74, 105, 111$
7	16	10	$n = 60, 120$
8	18	8	$n = 21, 26, 27, 39, \dots$

Table 4: The period-one construction reaches the ceiling $2k + 2$ for $k \leq 5$ and, in the computed range, falls well short for $k \geq 6$.

k	$2k + 2$	period d	$\lceil k/2 \rceil$	n	smallest required entry
6	14	3	3	60, 72	0.188, 0.162
7	16	4	4	96	0.107
8	18	4	4	96	0.260

Table 5: Period- d certificates attaining the ceiling for $k \geq 6$. The period used is in every case the one the rank law prescribes.

k	dock $(a_0, c_0, d_0, e_0), b_0 = 1$	ℓ_0	rank = $\dim Sp$	$\det B$
2	(0, 1, 0, 1)	15	21 = 21	1
3	(1, 1, 1, 1)	21	36 = 36	1
4	(0, 1, 1, 1)	29	55 = 55	1
5	(0, 1, 0, 1)	39	78 = 78	1
6	(0, 1, -1, 2)	49	105 = 105	4096
7	(0, 1, 0, -1)	61	136 = 136	1
8	(-1, 1, 0, 2)	75	171 = 171	65536
9	(0, 1, 0, 2)	89	210 = 210	262144

Table 6: The docks and lengths at which the hypothesis of Theorem 117 is certified. Generated by `verification/verify_all.py`.